

7	(a)	(i)	The central maximum is white in colour, while the first order maxima on either side is a spectrum. [B1] Central maximum appears white because all the components of the white light will meet constructively at the centre of the screen. OR The central maximum occurs due to the waves meeting in phase. [B1] For the first order maxima, based on the equation $d \sin \theta = \lambda$, different wavelengths of light will end up at different positions on the screen (diffracted to different angles), thereby forming a spectrum. [B1]	
		(ii)	In order for the full spectrum to be seen, red light must be captured on the screen. The visible wavelength occupies 400nm to 700nm, [B1] Evaluating at $\theta = 90^\circ$ and with $d = \frac{1 \times 10^{-3}}{400} = 2.5 \times 10^{-6} \text{ m}$ [M1] $d \sin \theta = n\lambda$ $\sin \theta < 1$ $\frac{n\lambda}{d} < 1$ $n < \frac{0.0000025}{700 \times 10^{-9}}$ < 3.57 Therefore, the maximum order is 3. [A1]	
	(b)	Interference is the phenomenon which occurs when two or more waves of the same type overlap (superpose) according to the Principle of Superposition. [A1]		
	(c)	(i)	1.	When frequency decreases, since v remains constant, λ will increase. [B1] From the formula, $x = \frac{\lambda D}{a}$, the distance between the maxima will increase. [B1] Hence Q, R, S, T, U will be further apart. [B1]

			2.	Increasing the amplitude causes the energy of the water waves to increase as well. [B1] The amplitudes at Q, S and U will be stronger while amplitudes at R, T will remain the same. [B1]
			3.	Since the dippers are no longer coherent or they do not have a constant phase relationship/ difference. [B1] Hence, there will not be any observable interference/ no maxima and minima observed at Q, R, S, T, U . [B1]

		(ii)	1.	$x = \frac{\lambda D}{a}$ $0.040 = \frac{\lambda \times (1.12)}{0.056}$ $\lambda = 0.0020 \text{ m} \quad [\text{C1}]$ $v = f\lambda$ $= 25 \times 0.0020 \quad [\text{C1}]$ $= 0.050 \text{ m s}^{-1} \quad [\text{A1}]$ Alternative method: Path difference = $D_2Q - D_1Q =$ $\sqrt{(1120^2 + 68^2)} - \sqrt{(1120^2 + 12^2)} = 1.998 \text{ mm} \quad [\text{C1}]$ Since Q is first order maxima, $\lambda = 1.998 \approx 2.00 \text{ mm}$ $v = f\lambda$ $= 25 \times 0.0020 \quad [\text{C1}]$ $= 0.050 \text{ m s}^{-1} \quad [\text{A1}]$	
			2.	$\Delta\phi = \frac{x}{\lambda} \times 2\pi$ $= \frac{0.0015}{0.0020} \times 2\pi \quad [\text{C1}]$ $= 1.5\pi \text{ or } 4.71 \quad [\text{A1}]$ correct unit: rad or ° [B1]	

8	(a)	(i)	Radioactive decay is the spontaneous emission of particles or radiation
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